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IBM AI Racing League Country Challenge 2026

Building an autonomous AI driver for the TORCS
simulator.

AiBC team: Wiktoria, Óscar, Mario, Krystian, Jakub, Dominik, Diego, Álvaro

Date: 17/03/2026



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IBM AI RACING LEAGUE - COUNTRY CHALLENGE 2026

Business Justification

1. Simulating Industry Realities

- Bridging the gap between academic theory and real-world tech industry demands
- Proving our ability to not only build complex AI solutions but to explain and defend them effectively

2. Innovation & Value

- Creating value by combining hard technical skills with essential soft skills
- Becoming complete technologist who are both technically proficient and communicatively strong

3. Industry Connection

- Utilizing the challenge as a platform to highlight our industry-readiness to IT experts on a global stage



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Project scope (goal/aims)

Primary Goal

To build and test the ultimate autonomous AI driver for the Corkscrew circuit.

Technical Aim

- Enviroment → Navigating the Corkscrew circuit from a standing start
- Constraints → Modifying strictly the AI Python code (internal TORCS car dynamics must remain unchanged)
- Deliverables → Achieving and recording the fastest possible lap time

Strategic Aim

Demonstrate the use of data-driven decision making in autonomous racing and the logic behind the speed



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Features to be implemented

Key Challenges

- High-speed corner handling on the Corkscrew section
- Maintaining optimal racing line
- Balancing acceleration and stability
- Avoiding track limits and collisions

Technical stack

- TORCS simulator
- Python / AI driver code
- Data analysis tools
- GitHub for version control





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Key Challenge



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State of the art

ORIGINS

IBM AI Racing League

- Launched by IBM as a global STEM & AI education initiative
- Uses TORCS (The Open Racing Car Simulator) as the competition environment
- Teams from universities worldwide compete to build the fastest autonomous driver

SIMULATOR

TORCS — The Environment

- Open-source physics-based racing simulator, used in AI research since 2013
- Provides 19 range sensors, speed, angle, and track position data
- Widely adopted in autonomous driving and RL research (SCRC benchmark)

COMPETITION FORMAT

How the Challenge Works

- Teams submit AI drivers that race on a fixed circuit (Corkscrew track)
- Judged on lap time AND ability to explain technical decisions clearly
- Introduces the 'Dual-Core Athlete': technical skill + communication

EXISTING APPROACHES

What Has Been Done

- PID controllers and rule-based bots achieve consistent, stable laps
- Reinforcement Learning (DQN, PPO) applied to TORCS by academic teams
- IBM Granite integration in racing context is a novel, emerging area

IBM GRANITE

Core AI Technology

- IBM Granite is the mandatory foundation model for all teams
- Used to optimize braking, acceleration, and racing logic
- Certified via IBM SkillsBuild — required for all participants

INDUSTRY RELEVANCE

Why It Matters

- Mirrors real-world autonomous vehicle decision-making challenges
- Global showcase connecting student talent with IBM industry experts
- Bridges AI engineering skills with professional communication standards



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Techniques & Technologies to be used



AI & MACHINE LEARNING

- **IBM Granite**
Core AI engine - model training & inference
- **IBM SkillsBuild**
Mandatory Granite certification platform
- **Reinforcement Learning**
Agent training & lap-time optimisation
- **Agent-Based Design**
Autonomous decision-making logic



DEV & SIMULATION

- **Python**
Primary language - driver control scripts
- **TORCS Simulator**
Physics-based racing environment
- **NumPy / Pandas**
Telemetry data processing & feature engineering
- **Corkscrew Track**
Official qualification environment



COLLABORATION & DOCS

- **GitHub**
Version control & project documentation
- **Trello**
Sprint tracking & task management
- **Messenger**
Allows team communication between all members
- **UML / Flowcharts**
System architecture & design documentation

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Team roles

DEVELOPERS

- Jakub Swatowski
- Wiktoria Wojdyło
- Dominik Pokrzywa
- Álvaro Puebla Ruisánchez
- Mario Alonso

TESTERS

- Diego Ceballos
- Óscar López

DEV OPS

- Krystian Tomczyk



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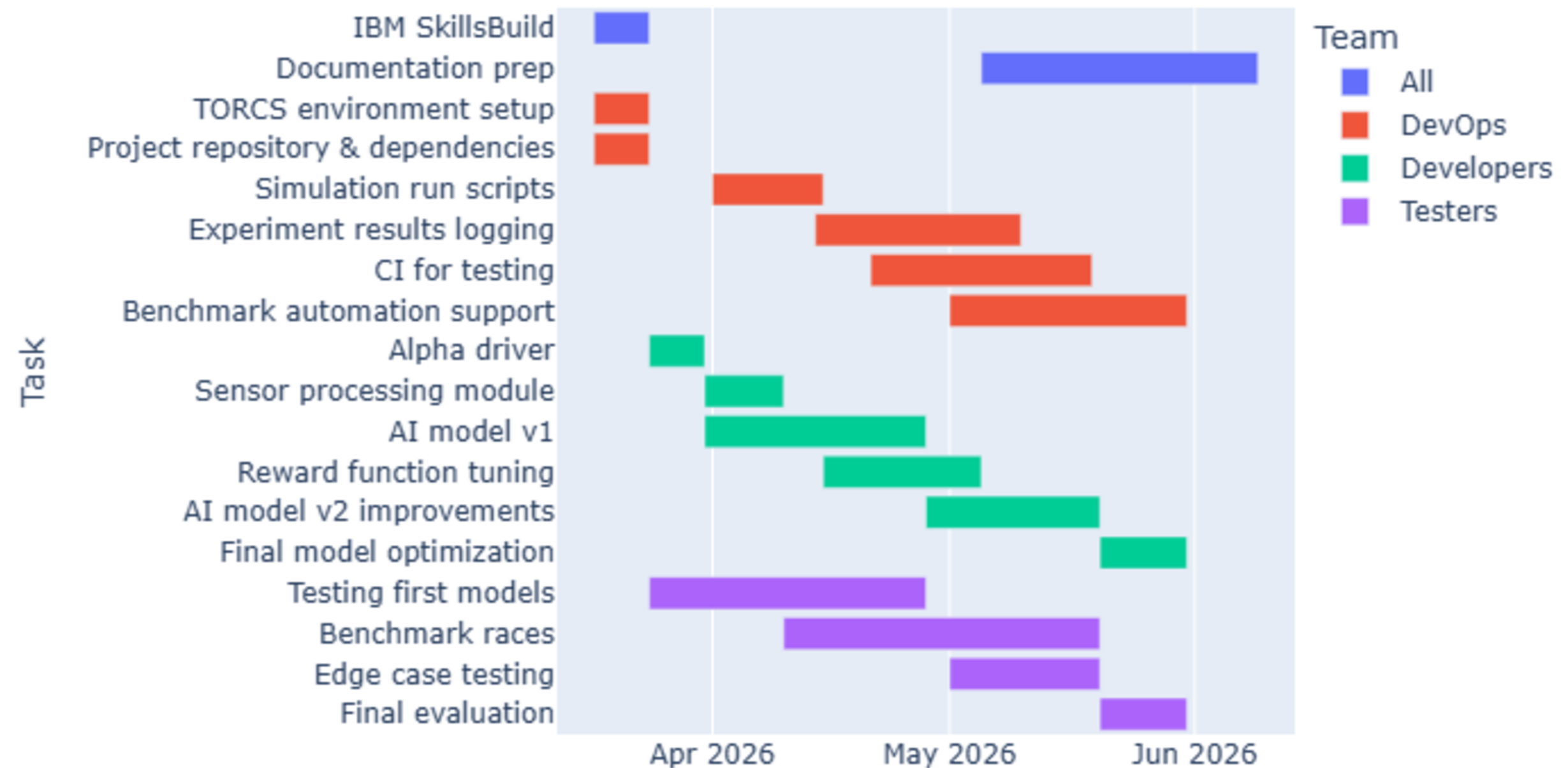


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Project plan with milestones

IBM AI Racing Project Plan





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Cost estimation

- Learning - free IBM SkillBuild courses.
- Technologies - free python libraries, TORCS.
- Project management - GitHub, Trello.
- Hardware - personal computers.



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LET'S GET TO WORK



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